



CAPSTONE PROJECT REGISTER

Class: _____ Duration time: from/.../.... To/... /.....

(*) Profession: < Artificial Intelligence > Specialty: CV ☐ NLP ☒ DS ☐ Other ☐

(*) Kinds of person make registers: Lecturer ☒ Students ☐

1. Register information for supervisor (if have)

	Full name	Phone	E-Mail	Title
Supervisor 1	Đặng Ngọc Minh Đức		ducdnm2@fe.edu.vn	Assoc.Prof.

2. Register information for students (if have)

	Full name	Student code	Phone	E-mail	Role in Group
Student 1					Leader
Student 2					Member
Student 3					Member

3. Register content of the Capstone Project

(*) 3.1. Capstone Project name:

English: Efficient Vietnamese Visual Question Answering using Vision-Language Models

Vietnamese: Nghiên cứu hệ thống hỏi đáp trực quan tiếng Việt hiệu quả sử dụng mô hình Vision-Language

Abbreviation: ViVQA-VLM

a. Context:

Visual Question Answering (VQA) is a multimodal AI task that requires a system to answer natural-language questions based on the visual information in an image. Unlike image classification or object detection, VQA demands a comprehensive understanding of visual content, language semantics, and reasoning. Recent Vision-Language Models (VLMs) — BLIP-2, Florence-2, LLaVA, and Qwen2.5-VL — have substantially improved multimodal reasoning by jointly learning visual and textual representations, showing strong zero-shot and few-shot capabilities. In parallel, Vietnamese datasets such as ViVQA and ViTextVQA have created new opportunities to develop intelligent multimodal systems for the Vietnamese language.

Research Gap:

- English-centric pretraining: most VLMs are trained predominantly on English data, so their visual-linguistic alignment is optimized for English rather than Vietnamese.
- Weak Vietnamese performance, especially with scene text: effectiveness on Vietnamese VQA — particularly text-in-image (OCR) questions and Vietnamese diacritics/code-switching — remains limited.
- PEFT for Vietnamese VLMs is under-explored: parameter-efficient fine-tuning (LoRA/QLoRA) of large VLMs for Vietnamese VQA has not been studied thoroughly.
- Hallucination: VLMs still produce fluent but incorrect answers (object/text hallucination), a major reliability challenge for Vietnamese VQA.

Therefore, efficiently adapting open-source VLMs to Vietnamese VQA under a strict parameter-efficient budget — rather than training from scratch or full fine-tuning — is a practical and underexplored research direction. This capstone targets a concrete, measurable technical contribution rather than an application demo.

b. Objectives:

The objectives of this capstone project are:

- Study the fundamentals of Visual Question Answering and Vision-Language Models.
- Study Vietnamese VQA benchmark datasets (ViVQA, ViTextVQA).
- Evaluate several pretrained VLMs on Vietnamese benchmarks under a parameter-efficient budget.
- Investigate lightweight adaptation techniques (PEFT — LoRA/QLoRA, prompting, OCR/retrieval augmentation).
- Analyze hallucination behavior on Vietnamese VQA.
- Develop a Vietnamese Visual Question Answering prototype.

Research objective (contribution):

Propose and experimentally validate at least one technical improvement over a reproduced VLM baseline (e.g., a PEFT recipe, OCR-enhanced prompting, or retrieval-augmented VQA) that measurably improves Vietnamese VQA accuracy or reduces hallucination, supported by ablation and error analysis, without full fine-tuning.

c. Proposed solution:

Dataset Preparation:

Investigate and preprocess Vietnamese benchmark datasets — ViVQA (general) and ViTextVQA (scene-text). Standardize image handling, question/answer normalization, OCR extraction (e.g., PaddleOCR / VietOCR) for text-in-image questions, and a fixed train/validation/test split.

Baseline Requirement:

Each group must reproduce exactly one open-source VLM baseline before any improvement, so the examination board has a clear reference point:

- BLIP-2.
- Florence-2.
- Qwen2.5-VL.

Required Improvement (Novelty):

After the baseline is reproduced, students must implement at least one of the following (or an equivalent contribution):

- LoRA / QLoRA fine-tuning.
- OCR-enhanced prompting (PaddleOCR / VietOCR).
- Retrieval-Augmented VQA (RAG).
- Prompt optimization.
- Multimodal prompt engineering.
- Lightweight VLM adaptation.

The improvement must be motivated by the Research Gap and evaluated against the reproduced baseline under identical conditions.

Training Constraint (mandatory):

Full fine-tuning is not permitted. Adaptation must use parameter-efficient fine-tuning only:

- PEFT (adapter / LoRA family).
- LoRA.
- QLoRA (4-bit or 8-bit quantized).

This keeps the project runnable on Google Colab Pro/Premium or the university GPU server, and makes efficiency an explicit design constraint.

Experimental Evaluation:

Primary metrics:

- Exact Match (EM).
- VQA Accuracy.

- ANLS (especially for ViTextVQA / scene-text questions).

Additional analysis:

- Hallucination analysis (rate and type of hallucinated objects/text).
- Human evaluation on a sample (if feasible), for answer correctness and faithfulness.

Note: BLEU/CIDEr are not used as primary metrics; they may be reported only as secondary indicators for open-ended generative answers.

Ablation study (required):

- Zero-shot vs. prompt-optimized vs. LoRA/QLoRA-adapted.
- With vs. without OCR-enhanced prompting.
- With vs. without retrieval augmentation.
- Different VLM backbones.

Error analysis (required):

- Error breakdown by question type (yes/no, counting, text-reading, reasoning).
- Failure-case visualization on text-heavy Vietnamese images.
- Hallucination case studies and Vietnamese-specific challenges (diacritics, code-switching, OCR errors).

Evaluation protocol (for consistency across groups):

Report the exact dataset split, random seed(s), hardware configuration, and inference (and PEFT fine-tuning) time.

Prototype Development:

A web-based prototype that uploads a Vietnamese image, accepts a question in Vietnamese, generates a natural-language answer, and visualizes model confidence (and, where relevant, the extracted OCR text / retrieved evidence).

d. Algorithm requirement

Training / Adaptation

- Datasets: ViVQA (general), ViTextVQA (scene-text).
- Models: one reproduced VLM baseline (BLIP-2 / Florence-2 / Qwen2.5-VL) adapted with PEFT only (LoRA / QLoRA 4-bit or 8-bit); no full fine-tuning.
- Supporting tools: PaddleOCR / VietOCR for text-in-image questions; optional retrieval index for RAG.

Testing

- Primary metrics: Exact Match, VQA Accuracy, ANLS.
- Also: hallucination analysis and (optional) human evaluation.
- Baseline comparison: reproduced baseline vs. the improved model; at least one additional VLM for reference.
- Ablation and error analysis are mandatory.
- Evaluation protocol: report split, seed, hardware, and inference/fine-tuning time.

Deployment

- Runtime environment: Colab Pro/Premium or a university GPU server; Python + PyTorch + HuggingFace Transformers + PEFT.
- Interface: a web prototype (image upload, Vietnamese Q&A, confidence/OCR visualization).

(*) 3.2. Main proposal content (including result and product)

a. Theory and practice (document):

Apply knowledge from AIL302m (AI), AIP391 (Machine Learning), DPL301m (Deep Learning), NLP, Computer Vision, and Software Engineering. Literature review: Vision-Language Models, Visual Question Answering, Multimodal Learning, Prompt Engineering, Parameter-Efficient Fine-Tuning (LoRA/QLoRA), OCR for VQA, Retrieval-Augmented Generation, hallucination in VLMs, and evaluation of multimodal AI systems.

- Artificial Intelligence technologies:

Framework: PyTorch, HuggingFace Transformers, PEFT/LoRA/QLoRA (bitsandbytes).

Foundation models: BLIP-2, Florence-2, LLaVA, Qwen2.5-VL.

OCR: PaddleOCR / VietOCR. Deployment: Python.

b. Products:

- A Vietnamese Visual Question Answering system (PEFT-adapted VLM).
- A web-based demonstration prototype.
- Evaluation reports comparing VLMs, with ablation studies and hallucination analysis.
- A complete capstone thesis; results encouraged toward undergraduate research publications.

Reproducibility package (required):

Students must provide:

- Source code.

- LoRA/QLoRA adapter weights.
- Configuration files.
- Inference script.
- README with setup and run instructions.
- Experiment log (metrics, seeds, prompts, settings per run).

Proposed Tasks (aligned with the AIP491 15-week schedule):

- **Task package 1 (Weeks 1–3):** Literature review, dataset exploration, environment setup, and selection of the target VLM baseline (→ Report 1: Introduction; Report 3: Existing Systems).
- **Task package 2 (Weeks 4–6):** Dataset preprocessing (incl. OCR) and reproduction of the chosen baseline (→ Report 4: Methodology).
- **Task package 3 (Weeks 7–10):** PEFT adaptation (LoRA/QLoRA), OCR/retrieval or prompt improvement (→ Report 4/5).
- **Task package 4 (Weeks 11–13):** Experimental evaluation, ablation studies, hallucination/error analysis, comparative analysis (→ Report 6 & 7: Results/Discussion).
- **Task package 5 (Weeks 14–15):** Prototype development, thesis writing, and final presentation (→ Final report and defense).

AI Ethics and Responsible AI:

Consistent with AIP491 CLO7 (Professionalism and Ethical Conduct in AI) and the AI Ethics and Data Privacy session:

- Privacy of uploaded images (faces, documents, personal data) and appropriate dataset licensing.
- Bias and fairness — English-centric or cultural biases in VLMs may disadvantage Vietnamese content or specific groups.
- Hallucination and reliability — fluent-but-wrong answers, and limits of automated VQA in sensitive contexts.
- Responsible use — clear communication of confidence and system limitations.

Expected Learning Outcomes:

Upon completion, students will be able to (mapped to AIP491 CLOs):

- Build multimodal AI systems using vision-language foundation models (CLO3, CLO8).

- Reproduce a VLM baseline and propose and validate a PEFT-based improvement (CLO3, CLO8, CLO10).
- Conduct reproducible experiments with proper protocol, ablation, and error/hallucination analysis (CLO2, CLO5, CLO9).
- Critically analyze results and Vietnamese-specific failure modes (CLO9).
- Develop and deploy an interactive AI prototype (CLO3, CLO6).
- Apply professional and ethical standards to multimodal systems (CLO7).
- Communicate research findings effectively in a written thesis and oral defense (CLO5, CLO6).

References:

Foundations

- [1] J. Li, et al. BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models. ICML, 2023.
- [2] H. Liu, et al. Visual Instruction Tuning (LLaVA). NeurIPS, 2023.

Benchmark / Dataset

- [3] K. Q. Tran, et al. ViVQA: Vietnamese Visual Question Answering. PACLIC, 2021.
- [4] Q. V. Nguyen, et al. ViTextVQA: A Large-Scale VQA Dataset for Evaluating Vietnamese Text Comprehension in Images. arXiv:2404.10652, 2024 (extended version in Expert Systems with Applications, 2025).

Baseline Models

- [5] B. Xiao, et al. Florence-2: Advancing a Unified Representation for a Variety of Vision Tasks. CVPR, 2024.
- [6] S. Bai, et al. Qwen2.5-VL Technical Report. arXiv, 2025.

Recent Representative Methods

- [7] N. S. Nguyen, et al. Advancing Vietnamese Visual Question Answering with Transformer and Convolutional Integration. Computers & Electrical Engineering, vol. 119, 2024.
- [8] J. Chen, et al. Florence-VL: Enhancing Vision-Language Models with Generative Vision Encoder and Depth-Breadth Fusion. CVPR, 2025.
- [9] Y. Li, et al. Evaluating Object Hallucination in Large Vision-Language Models (POPE). EMNLP, 2023.

Survey

- [10] J. Zhang, et al. Vision-Language Models for Vision Tasks: A Survey. IEEE TPAMI, 2024.

4. Other comments (propose all relevant things if any)

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....., date / /

Supervisor (If have)
(Sign and full name)

On behalf of the Registers
(Sign and full name)